SISO Design Limitations – Zeros/Poles

(SISO系统设计的局限性 – 零极点)

General Effect of Open Loop Poles/Zeros

- **Lemma 3(引理3)**

For *stable* CL system with poles located to the left of $-\alpha$ for some $\alpha>0$, assume that the <u>controller has at least one pole at origin</u>. Then, for any uncancelled plant zero z_0 or pole p_0 that is to the right of CL poles (i.e., $R\{z_0\}>-\alpha$ or $R\{p_0\}>-\alpha$), we have（对于极点在$-\alpha(\alpha>0)$左侧的稳定闭环系统，假设控制器在原点处至少有一个极点。那么，对任何闭环极点右侧未消掉的系统零点或极点(也就是$R\{z_0\}>-\alpha$ 或$R\{p_0\}>-\alpha$)，我们有

(i) For a positive unit reference step or a negative unit step output disturbance, error $e(t)$ satisfies(对于正单位参考阶跃或负单位阶跃输出干扰，误差$e(t)$满足)

$$\int_0^\infty e(t)e^{-z_0 t}dt = \frac{1}{z_0}$$

$$\int_0^\infty e(t)e^{-p_0 t}dt = 0$$

$$E(s) = S(s)\underbrace{[R(s)-d_0(s)]}_{1/s} = S(s)\frac{1}{s}$$

At $z_0, S(z_0)=1$ (zeros of L(s))

According to definition of Laplase transform (根据拉普拉斯变换的定义)

$$\int_0^\infty e(t)e^{-z_0 t}dt = E(z_0) = S(z_0)\frac{1}{z_0} = \frac{1}{z_0}$$

At $p_0, S(p_0)=0$ (poles of L(s))

$$\int_0^\infty e(t)e^{-p_0 t}dt = E(p_0) = 0$$

General Effect of Open Loop Poles/Zeros

(ii) For a positive unit step reference and for z_0 in RHP (unstable zero) (对于正的单位阶跃参考和右半平面中的z_0(不稳定零点)),

$$\int_0^\infty y(t)e^{-z_0 t}dt = 0$$

(iii) For a negative unit step input disturbance (对于负的单位阶跃输入干扰),

$$\int_0^\infty e(t)e^{-z_0 t}dt = 0$$

$$\int_0^\infty e(t)e^{-p_0 t}dt = \frac{D_c(p_0)}{p_0 N_c(p_0)}$$

General Effect of Open Loop Poles/Zeros

- **Observations(观察)**

(i) A real *stable* (LHP) *zero* to the right of all CL poles produces *overshoot* in the step response(在所有闭环极点右侧的实稳定零点(LHP)会在阶跃响应中产生超调)

$$\left.\begin{array}{l}\Rightarrow \int_0^\infty e(t)e^{\alpha t}dt = -\dfrac{1}{\alpha} < 0 \\ \text{since } e(t) > 0 \text{ at least at the beginning}\end{array}\right\} \Rightarrow$$

$$z_0 = -\alpha, \alpha > 0$$

$e(t)$ has to negative during time intervals!
($e(t)$在这段时间内必须为负)

Small stable zero leads to large overshoot!
(小的稳定零点会导致大的超调)

(ii) A real *unstable* (RHP) *zero* always produces *undershoot* in the step response. The amount of undershoot grows as the zero approaches origin(实不稳定(RHP)零点总是在阶跃响应中产生负调。当零点接近原点时，负调量增加)

$$\left.\begin{array}{l}\int_0^\infty y(t)e^{-\alpha t}dt = 0, \quad \text{since } y(t) = 1 \text{ for } \forall t > t_s \\ \int_0^{t_s} y(t)e^{-\alpha t}dt + \int_{t_s}^\infty e^{-\alpha t}dt = 0, \Rightarrow \int_0^{t_s} y(t)e^{-\alpha t}dt + \dfrac{1}{\alpha}e^{-\alpha t_s} = 0\end{array}\right\} \Rightarrow$$

$$z_0 = \alpha, \alpha > 0$$

$y(t)$ has to negative during time intervals when $t<t_s$!
（在$t<t_s$这段时间内必须为负）

Small unstable zero leads to large undershoot!
（小的不稳定零点会导致大的负调）

General Effect of Open Loop Poles/Zeros

- **Observations(观察)**

(iii) Any *real* open-loop *pole* to the right of all CL poles will produce *overshoot* in a one-DOF control architecture (在单自由度控制结构中，所有闭环极点右侧的任何实开环极点都会产生超调)

$$\int_0^{\infty} e(t)e^{-p_0 t}dt = 0$$

$$\Rightarrow$$

For fast unstable open-loop poles (对快速不稳定开环极点来说): $p_0 = \alpha,\ \alpha >> 1$

$e^{-p_0 t}$ drops zero very quickly. ($e^{-p_0 t}$ 迅速降为0)

General Effect of Open Loop Poles/Zeros

- **Guidelines(指南)**

(i) The *CL bandwidth* should be in practice be set *less* than the *smallest unstable zero* to avoid large undershoot in step response **(在实际应用中，闭环带宽应小于最小的不稳定零点，以避免阶跃响应中出现较大的负调)**

(ii) It is advisable to set CL bandwidth *greater* than *real part* of any *unstable pole* to avoid large overshoot**(建议闭环带宽大于任何不稳定极点的实部，以避免较大的超调)**

(iii) To avoid overshoot produced by *stable* plant or controller zeros, cancel them outside the loop in a two-DOF design using poles of filter TF $F(s)$**(为了避免系统或控制器的稳定零点产生的超调，在双自由度设计的闭环回路之外设计滤波器, 其传递函数是$F(s)$,用$F(s)$的极点来消去稳定零点)**

Examples - Open Loop Poles/Zeros

- **System**

$$P(s)=\frac{s-z_p}{s(s-p_p)} \qquad C(s)=\frac{K_c(s-z_c)}{s-p_c}$$

Desired CL Poles (期望闭环极点):

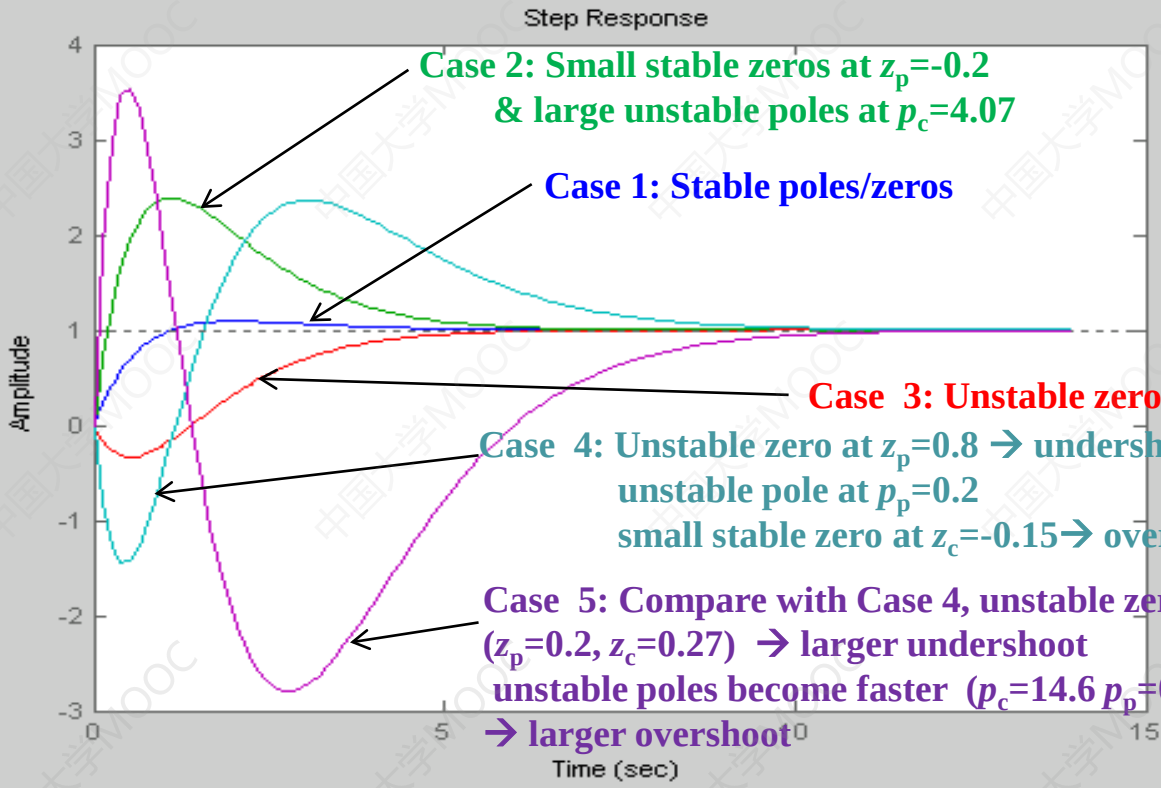
$$p_{CL1d}=p_{CL2d}=p_{CL3d}=-1$$

Target CL bandwidth is approximately 1 rad/s! **(目标闭环带宽约为1rad/s)**

$$K_c=\frac{3+1/z_p+3p_p+p_p^{\,2}}{p_p-z_p},\quad z_c=\frac{1}{K_c z_p},\quad p_c=K_c-3-p_p$$

	Case 1	Case 2	Case 3	Case 4	Case 5
	$p_p=-0.2$ $z_p=-0.8$	$p_p=-0.8$ $z_p=-0.2$	$p_p=-0.8$ $z_p=0.8$	$p_p=0.2$ $z_p=0.8$	$p_p=0.8$ $z_p=0.2$
K_c	1.98	6.27	−1.56	−8.15	18.4
p_c	−0.82	4.07	−3.76	−11.4	14.6
z_c	−0.63	−0.8	−0.8	−0.15	0.27

Examples - Open Loop Poles/Zeros



Case 2: Small stable zeros at z_p=-0.2
& large unstable poles at p_c=4.07

Case 1: Stable poles/zeros

Case 3: Unstable zero at z_p=0.8

Case 4: Unstable zero at z_p=0.8 → undershoot
unstable pole at p_p=0.2
small stable zero at z_c=-0.15→ overshoot

Case 5: Compare with Case 4, unstable zeros become smaller
(z_p=0.2, z_c=0.27) → larger undershoot
unstable poles become faster (p_c=14.6 p_p=0.8)
→ larger overshoot

Effect of Imaginary Poles/Zeros

- **Corollary 1(推论1)**

Consider the *stable* CL system in Lemma 3. Then, for a unit step reference input (考虑引理3中的稳定闭环系统，那么对单位阶跃参考输入)

(i) If plant has a pair of zeros at $\pm j\omega_0$, then(如果系统有一对共轭零点)

$$\int_0^\infty e(t)\cos(\omega_0 t)\,dt = 0$$

$$\int_0^\infty e(t)\sin(\omega_0 t)\,dt = \frac{1}{\omega_0}$$

$$\Leftarrow \int_0^\infty e(t)e^{-j\omega_0 t}dt = \frac{1}{j\omega_0} = -\frac{1}{\omega_0}j$$

(ii) If plant has a pair of poles at $\pm j\omega_0$, then (如果系统有一对共轭极点)

$$\int_0^\infty e(t)\cos(\omega_0 t)\,dt = 0$$

$$\int_0^\infty e(t)\sin(\omega_0 t)\,dt = 0$$

$$\Leftarrow \int_0^\infty e(t)e^{-j\omega_0 t}dt = 0$$

Effect of Slow Imaginary Zeros

- **Example 1**

Assume ω_0 is small when compared to CL bandwidth (假设ω_0小于闭环带宽):

$$t_s \le \frac{\pi}{\omega_0}$$

Since error e(t) becomes very small after the settling time (由于在调整时间之后的阶段误差非常小)

$$\int_{t_s}^{\infty} e(t)\cos(\omega_0 t)dt = 0, \int_{t_s}^{\infty} e(t)\sin(\omega_0 t)dt = 0$$

Estimated Error Bound from Corollary 1(从推论1估计误差范围):

$$\frac{1}{\omega_0} = \int_0^{\infty} e(t)\sin(\omega_0 t)dt \approx \int_0^{t_s} e(t) \underbrace{\sin(\omega_0 t)}_{\ge 0,\ \text{with } \omega_t \in [0,\pi], \forall t \le t_s} dt \le e_{\max}\int_0^{t_s}\sin(\omega_0 t)dt \le e_{\max}\frac{1-\cos(\omega_0 t)}{\omega_0}$$

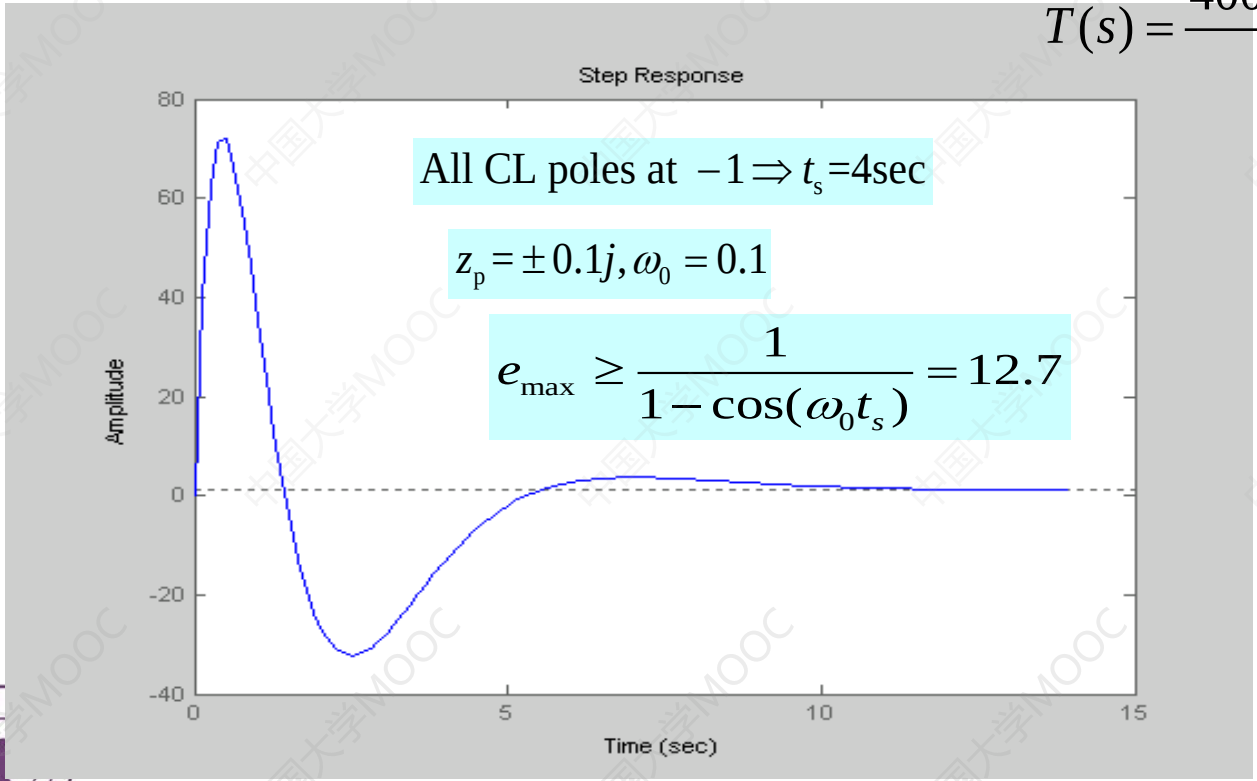
$$\Rightarrow \quad e_{\max} \ge \frac{1}{1-\cos(\omega_0 t_s)} \quad \to \quad \infty \quad \text{as} \quad \omega_0 t_s \to 0$$

Maximum error will be very large if one tries to make CL bandwidth greater than the frequency of resonant zeros (or imaginary zeros) (如果试图使闭环带宽大于谐振零点(或虚轴上零点)的频率，则控制系统的最大误差将会非常大)

Example of Slow Imaginary Zeros

- **Example 2**

$$T(s)=\frac{400(s+1)(s+0.25)(s^2+0.01)}{(s+1)^5}$$



All CL poles at $-1 \Rightarrow t_s = 4\text{sec}$

$z_p = \pm 0.1j, \omega_0 = 0.1$

$$e_{\max} \geq \frac{1}{1-\cos(\omega_0 t_s)} = 12.7$$

$$P(s)=\frac{100s^2+1}{s^2(s+1)},$$

$$C(s)=\frac{(s+1)(4s+1)}{s^2-396s-94}$$

$\Rightarrow p_{1c} = -0.24, p_{2c} = 396$

Note huge maximum error during transient response !(注意瞬态响应过程中很大的误差)

Summary

- **Fundamental Structural Limitations (基本结构的限制)**
  - Poles in open right half plane (ORHP) or Unstable Poles **(开右半平面(ORHP)的极点或不稳定极点)**
  - Zeros in ORHP or Unstable Zeros **(ORHP的零点或不稳定零点)**
  - Stable Zeros but close to Origin **(稳定但接近原点的零点)**
  - Poles on Imaginary Axis **(虚轴上的极点)**
  - Zeros on Imaginary Axis **(虚轴上的零点)**